DRUMPANEL · REGULATORY REGISTER

DrumPanel regulatory and standards register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/07-regulatory.md @ de21d74		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First register (kit 1.8 upgrade of a TRL 3 repo); container preparation, hot work, flash point, machinery safety, ergonomics, gas detector and waste oil rows; Haiti rows to assess

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The bench set is a hand-powered tool published as an open engineering reference, never as certified equipment. Its design follows the container preparation and machinery safety practice that applies to hot and cold work on used drums; the national rules of the country where a set is first used (Haiti or wherever the partner cooperative works) are still to assess, above all for purge water and waste oil. Not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. Machinery and ergonomics rows shared across the Circular Materials focus area are cited from CRP-REG-001 (CoirPress, checked 2026-10-24). Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	29 CFR 1926.352(i) and 29 CFR 1910.252(a)(3), welding and cutting of used containers	United States	Drums that held flammable or toxic substances are filled with water or cleaned, ventilated and tested before welding, cutting or heating	The purge (water fill, drain) and the vapour check follow this practice although the bench set never heats a drum (PR-001, PR-002)	Design lead	Applies (as guidance outside the United States)
G2	AWS F4.1:2017, safe practices for the preparation of containers and piping for welding, cutting and allied processes	International practice	Cleaning by water, steam or chemicals, inerting, testing before work	Water cleaning is the method chosen (DMP-DDR-001, D1); the procedure sheets follow its sequence	Design lead	Applies (as guidance); text to read before TRL 4
G3	29 CFR 1915.14(b), hot work: no hot work at or above 10 % of the lower explosive limit	United States (shipyards)	The vapour threshold for hot work	The basis of PR-002 (flag at 10 % LEL); the bench set passes a drum only below 5 % LEL, half that	Design lead	Met by design
G4	Globally Harmonized System (GHS) classification of flammable liquids	International	Flash point categories: category 3 up to 60 °C, category 4 above 60 to 93 °C	The drum acceptance rule (PR-013) takes oils with flash points above 60 °C; whether to raise it to 93 °C is open decision 2 in DMP-DEC-001	Design lead	Applies
G5	IEC 60079-29-1, explosive atmospheres: performance requirements of detectors for flammable gases	International	Performance of portable flammable gas detectors	The bought pumped LEL detector (BOM line 4) should meet it, and be certified for use in hazardous areas	Partner (purchaser)	To assess when the detector is chosen
G6	ISO 12100, safety of machinery: risk assessment and risk reduction	International	Hazard identification and the order of risk reduction	DMP-RSK-001 follows its method for the shears, rolls, punch and chains	Design lead	Applies (as guidance for a non-certified hand tool)
G7	ISO 13857:2019, safety distances to prevent hazard zones being reached by upper and lower limbs	International	Openings in guards and their distance from danger points	Every slip roll and shear head slot is 8 mm or less and at least 45 mm from its nip (PR-010, DMP-CAL-001 section 7)	Design lead	Met by design on paper; guards as built checked at TRL 4

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G8	ISO 11228-1 (manual handling, lifting) and EN 1005-3 (recommended force limits for machinery operation)	International; Europe	Lifting masses and hand forces	Lifts against the 25 kg reference mass (PR-011); crank and lever forces at or below 150 N (PR-006), an EN 1005-3 assessment at TRL 4	Design lead	To assess (EN 1005-3 at TRL 4)
G9	EU Machinery Directive 2006/42/EC and Machinery Regulation (EU) 2023/1230	European Union	Machinery placed on the EU market	Machinery is defined as having a drive system other than directly applied human or animal effort (CRP-REG-001); the bench set is worked by hand, so it is probably outside the scope	Design lead	To assess (probably does not apply)
G10	EN 388, protective gloves against mechanical risks	Europe and international buyers	Cut resistance levels of gloves	BOM line 24 asks for EN 388 cut level C or higher for every task with cut edges	Partner (purchaser)	Applies
G11	National rules on waste oil and oily water	Country of use (Haiti first)	Storage, disposal and recycling of used oil and wash water	Purge water is settled and reused; skimmed oil goes to a closed container for a recycler (PR-014); the disposal route must be agreed with the partner	Partner	To assess
G12	National occupational safety rules for workshops	Country of use (Haiti first)	Workshop safety, young workers, protective equipment	Apprentices do this stage of the work today; age and training rules to check with the partner	Partner	To assess

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- 29 CFR 1926.352, Fire prevention, paragraph (i), Legal Information Institute, https://www.law.cornell.edu/cfr/text/29/1926.352, checked 2026-10-25.
- 29 CFR 1910.252, Welding, cutting and brazing, paragraph (a)(3), Legal Information Institute, https://www.law.cornell.edu/cfr/text/29/1910.252, checked 2026-10-25.
- 29 CFR 1915.14, Hot work, paragraph (b), Legal Information Institute, https://www.law.cornell.edu/cfr/text/29/1915.14, checked 2026-10-25.

- American Welding Society, F4.1:2017, https://pubs.aws.org/p/1759/f412017-safe-practices-for-the-preparation-of-containers-and-piping-for-welding-cutting-and-allied-processes, checked 2026-10-25.
- Workplace Health and Safety Queensland, "Flammable and Combustible Liquids" (GHS categories), https://worksafe.qld.gov.au/safety-and-prevention/hazards/hazardous-chemicals/specific-hazardous-chemicals/flammable-and-combustible-liquids, checked 2026-10-25.
- ISO 13857:2019, https://www.iso.org/standard/69569.html, checked 2026-10-25.
- CRP-REG-001 v0.2 (CoirPress, Circular Materials focus area): EU machinery definition, ISO 12100, ISO 13857, EN 1005-3, ISO 11228-1, checked 2026-10-24.
- IEC 60079-29-1 and EN 388: titles as published; clauses to be read with a qualified reviewer before any compliance claim.